STATS I · WEEK 2

Stats I week 2 formulas: Describing categorical data

Describing categorical data. Save this PDF for revision.

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Describing categorical data

Relative frequency

f/n

All relative frequencies add to 1.

Percentage

$f/n \times 100$

Pie-chart angle

$f/n \times 360°$

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Describing categorical data

Count from a slice

n × angle/360° or n × percentage

Mode

the category with the highest frequency

Median (ordinal only)

the middle category after ordering

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Charts

Chart	Rule
Bar chart	One bar per category, gaps between bars, height = frequency
Pie chart	Parts of a whole; slices add to 100% (360°)
Pareto chart	Bars in descending order plus a cumulative-percentage line
Stacked / side-by-side bars	Two categorical variables together (week 4)

Common traps

⚠ Mean makes no sense for categorical data, even when categories are coded as numbers.

⚠ Median needs order: fine for ordinal, meaningless for nominal.

⚠ Data can have more than one mode.

⚠ Misleading graphs: y-axis not starting at 0, 3-D pies, bars of unequal width.

Notes for every week of Stats I, free.

Full notes, the formula sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/stats-1 →

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